

F24-050-D-Scholarly

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Session 2021-2025

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June, 2025

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Chapter 1

Introduction

The purpose of this project proposal document is to outline the development of “Scholarly”, an AI-driven Smart Learning app that helps students overcome the difficulty of staying focused while studying via textbooks or long video lectures. This document aims to provide a comprehensive overview of the problem, explain the scope and the modules included in this project, highlight the users and the requirements and provide insight into the architecture and working of the product using UML diagrams. The document is intended for stakeholders, developers and project managers.

1.1 Problem Statement

According to research conducted by the American Physiology Journal [1], an average human being has an attention span of 15-20 minutes whilst watching an hour-long video but this drastically decreases to only 5-8 minutes when reading or learning via a book or any other form of textual content. In the modern educational era, students often struggle to maintain focus whilst watching long video lectures or learning via a textbook. The vast amount of information presented in lengthy video lectures can also lead to cognitive overload, making it harder for students to retain key concepts and stay engaged throughout the video lectures. This issue is worsened in subjects that require deep understanding, where consistent focus is necessary for understanding complex topics. Our research indicates that every 3 out of 4 students not only experience the issue of retaining focus while studying but are also in need of a better alternative solution that provides more concise and engaging learning materials to not only better understand complex topics with great speed and accuracy but also cater to students’ attention spans.

1.2 Scope

The project aims to develop an innovative educational app to revolutionize how students comprehend and interact with learning-based content. The app will enable its users to upload a book and automatically generate a personalized learning roadmap based on the uploaded content. The generated roadmap will act as a study plan for the user and will allow the user to track their progress as they move forward. For each topic in the generated roadmap, the app will provide detailed explanations for learning in the form of a short video format known as reels. The main objective of these reels will be to make complex topics easier to understand and learn them with great speed and accuracy. The app will also feature dedicated user settings and a dashboard for personalization and user data management enabling users to view their analytics and recommendations for the learning roadmap. The main objective behind this project is to enhance students' learning experience by combining traditional study methods with cutting-edge AI technology, making education more interactive, personalized, and efficient.

1.3 Modules

1.3.1 User Management

This module will focus on the development of a prototype, enabling a secure login and registration process. After logging in, the personalized dashboard will allow users to monitor their learning progress, access uploaded materials, and engage with other app features.

1. User Authentication
2. User Profile
3. User Settings
4. Personalized Dashboard

1.3.2 Roadmap Generation

This module will add the functionality of Book Upload that will act as an input for the system and will automatically generate a personalized learning roadmap for the user. The generated roadmap will act as a structured study plan for the user and will visually present the topics and subtopics in a chronological order based on the uploaded content.

1. Book Upload
2. Generate Roadmap
3. View Roadmap

1.3.3 Reels Generation

This module will focus on the development of features such as text summarization, which will enable the user to generate short text explanations for the topics and subtopics in the generated roadmap. This module will also feature the development of short videos (reels) for the roadmap, allowing the user to understand the content in a visual representation.

1. Generate Textual Summaries
2. Generate Reels
3. View Reels

1.4 User Classes and Characteristics

User class	Description
Students	Students are the primary users of the "Scholarly" app, who want to upload learning materials, generate study roadmaps, and engage with learning content through short videos (reels) and textual summaries. Students may upload multiple books or study materials for different subjects and will rely on the app to help them manage their learning progress across various topics. They will also use the app to break down complex topics into smaller, more manageable sections, relying on the app's personalized roadmaps and reels to focus on specific areas of study.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use Cases

2.1.1 Diagram

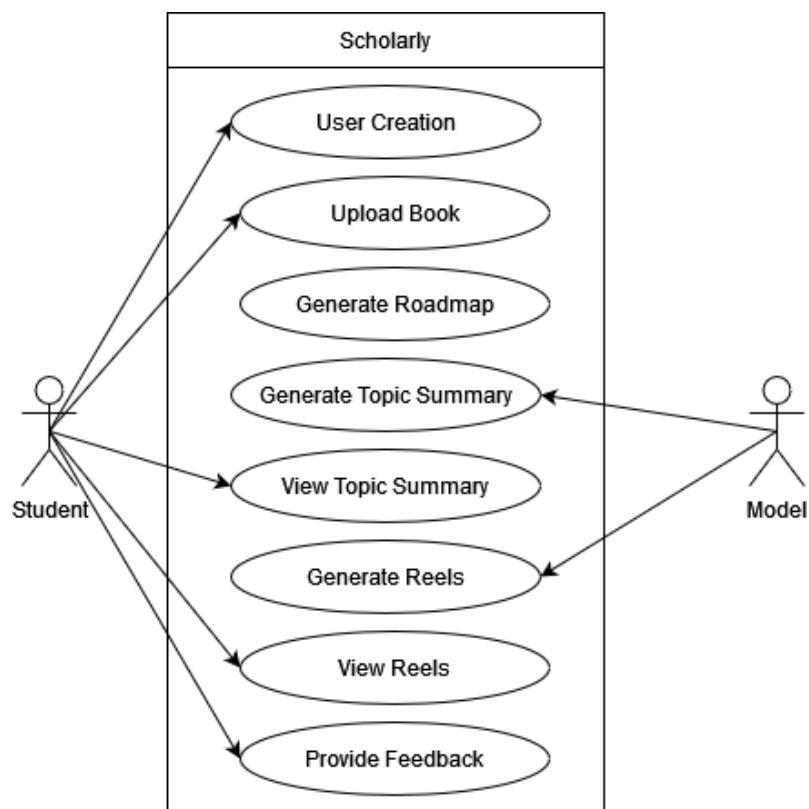


Figure 2.1: Use Case Diagram

2.1.2 User Creation

Name	User Creation	
Scope	Scholarly	
Level	User Level	
Primary Actor	Student	
Stakeholders and Interests	Student – Interested in creating a new user account. System – Interested in providing account creation and authentication service to the student.	
Preconditions	Student does not have a user account before. Student has all the valid information available such as Email and Password etc.	
Postconditions	Student has successfully created a new user account and logged into the system.	
Main Success Scenario	Actor Action	System Response
	Student desires to create a new user account.	
	Student provides the necessary information such as Email, Username, Password etc.	
		System checks if the information provided by the user is correct or not.
		System creates a new user account and updates the database.
		System logs in the new user.
Extensions	In case of any system error or any technical issue, the student may not be able to register an account. Student does not have the necessary information available, or the user has submitted wrong information. In both cases, the user cannot have a registered account.	

Table 2.1: User Creation

2.1.3 Upload Book

Name	Upload Book	
Scope	Scholarly	
Level	User Level	
Primary Actor	Student	
Stakeholders and Interests	Student – Interested in uploading their course book to generate a roadmap. System – Interested in getting the uploaded course book and updating the database.	
Preconditions	Student has a registered account and has logged in to the system.	
Postconditions	The system updates the database and redirects to the “Roadmap Page”.	
Main Success Scenario	Actor Action	System Response
	Student enters relevant information to log in.	
		System authenticates the information and grants access to the student.
	Student clicks on the “Books page”.	
		System redirects the student to the “Books page”.
	Student clicks on the “Upload New Book” button.	
		System displays an upload pop-up.
	Student uploads the book.	
		System preprocesses the book and updates the database.
		System redirects the student to the “Roadmap Page”.
Extensions	In case of any system error or technical issue, the student may not be able to upload their course book. In case of any system error or technical issue, the system may not be able to update the database or redirect to the “Roadmap Page”. The student does not remember their login credentials.	

Table 2.2: Upload Book

2.1.4 Generate Roadmap

Name	Generate Roadmap	
Scope	Scholarly	
Level	User Level	
Primary Actor	Student	
Stakeholders and Interests	Student – Interested in generating a roadmap for the uploaded course book. System – Interested in providing a roadmap to the student for the uploaded course book.	
Preconditions	Student has a registered account and has logged in to the system. Student has uploaded their course book.	
Postconditions	Student can view the generated roadmap for their course book.	
Main Success Scenario	Actor Action	System Response
	Student enters relevant information to log in.	
		System authenticates information and grants access to the student.
	Student clicks on “Books page”.	
		System redirects the student to the “Books page”.
	Student clicks on an uploaded book.	
		System retrieves the book contents from the database.
		System generates and displays the roadmap to the student.
Extensions	In case of any system error or technical issue, the system may not be able to generate and display the roadmap. In case of any system error or technical issue, the student may not be able to view their generated roadmap. The student does not remember their login credentials.	

Table 2.3: Generate Roadmap

2.1.5 Generate Topic Summary

Name	Generate Topic Summary	
Scope	Scholarly	
Level	System Level	
Primary Actor	System	
Stakeholders and Interests	System – Interested in generating a text summary for the student based on a selected topic.	
Preconditions	System has generated a roadmap and a topic has been selected by the student.	
Postconditions	System updates the database and stores the generated topic summary.	
Main Success Scenario	Actor Action	System Response
	Student selects a topic from the generated roadmap.	
		System fetches the description from the database based on the selected topic.
		System generates a summary based on the topic description.
		System updates the database and stores the generated summary.
Extensions	In case of any system error or technical issue, the system might not be able to generate a summary based on a selected topic.	

Table 2.4: Generate Topic Summary

2.1.6 View Topic Summary

Name	View Topic Summary	
Scope	Scholarly	
Level	User Level	
Primary Actor	Student	
Stakeholders and Interests	Student – Interested in viewing a summary of their selected topic. System – Interested in providing a text summary to the student based on a selected topic.	
Preconditions	Student has a registered account and has logged in to the system. Student has uploaded their course book and a roadmap has been generated.	
Postconditions	Student can view the topic summary of their selected topic.	
Main Success Scenario	Actor Action	System Response
	Student enters relevant information to log in.	
		System authenticates information and grants access to the student.
	Student clicks on “Books page”.	
		System redirects the student to the “Books page”.
	Student clicks on the roadmap.	
		System redirects the student to the “Roadmap Page”.
	Student selects a topic and clicks on “View Textual Explanation” button.	
		System displays the topic summary to the student.
Extensions	In case of any system error or technical issue, the student may not be able to view the topic summary. The student does not remember their login credentials.	

Table 2.5: View Topic Summar

2.1.7 Generate Reels

Name	Generate Reels	
Scope	Scholarly	
Level	User Level	
Primary Actor	Student	
Stakeholders and Interests	Student – Interested in generating reels of their selected topic. System – Interested in displaying the generated reels to the student based on a selected topic.	
Preconditions	Student has a registered account and has logged in to the system. Student has uploaded their course book and a roadmap has been generated.	
Postconditions	The system updates the database and stores the generated reels.	
Main Success Scenario	Actor Action	System Response
	Student logs into the system.	
	Student clicks on “Books page”.	
		System redirects the student to the “Books page”.
	Student clicks on the roadmap.	
		System redirects the student to the “Roadmap Page”.
	Student selects a topic and clicks on the “Generate Reels” button.	
		System redirects to the reels page.
		System fetches the contents from the database and generates reels based on the selected topic and description.
Extensions	In case of any system error or technical issue, the system may not be able to generate and display the reels. The student does not remember their login credentials.	

Table 2.6: Generate Reels

2.1.8 View Reels

Name	View Reels	
Scope	Scholarly	
Level	User Level	
Primary Actor	Student	
Stakeholders and Interests	Student – Interested in viewing reels of their selected topic. System – Interested in displaying reels to the student based on their selected topic.	
Preconditions	Student has a registered account and has logged in to the system. Student has uploaded their course book and a roadmap has been generated.	
Postconditions	System displays the reel to the student.	
Main Success Scenario	Actor Action	System Response
	Student enters relevant information to log in.	
		System authenticates the information and grants access to the student.
	Student clicks on “Books page”.	
		System redirects the student to the “Books page”.
	Student clicks on the roadmap.	
		System redirects the student to the “Roadmap Page”.
	Student selects a topic and clicks on the “View Reels” button.	
		System fetches the contents from the database and displays the reels to the student.
Extensions	In case of any system error or technical issue, the student may not be able to view the reels. The student does not remember their login credentials.	

Table 2.7: View Reels

2.1.9 Provide Feedback

Name	Provide Feedback
Scope	Scholarly
Level	User Level
Primary Actor	Student
Stakeholders and Interests	Student – Interested in providing feedback related to the generated reel to better optimize the system performance. System – Interested in recording the feedback and storing it in the database, and performing further actions accordingly.
Preconditions	Student has a registered account and has logged in to the system. System has generated the reel for the particular topic. Student has watched the reel of a particular topic for a fixed time.
Postconditions	The system updates the database and records the student's feedback.

Main Success Scenario	Actor Action	System Response
	Student enters relevant information to log in.	
		System authenticates the information and grants access to the student.
	Student clicks on “Books page”.	
		System redirects the student to the “Books page”.
	Student clicks on the roadmap.	
		System redirects the student to the “Roadmap Page”.
	Student selects a topic and clicks on the “View Reels” button.	
		System redirects to the reels page and displays the reels based on the selected topic.
	Student watches a fixed portion of the reel.	
		System tracks the student’s interaction with the reel.
	Student clicks on the “Feedback Button” and provides feedback.	
		System stores the feedback and updates the database.
Extensions	In case of any system error or technical issue, the student may not be able to provide feedback for a viewed reel. In case of any system error or technical issue, the system may not be able to update the database and store the provided feedback. The student does not remember their login credentials.	

Table 2.8: Provide Feedback

2.2 Functional Requirements

2.2.1 User Management

- FR001: The student should be able to create a new account by providing essential data such as email, username, password, and personal details.
- FR002: The system shall allow students to log in and authenticate them as users.
- FR003: The system shall offer an interface for users to view their profile.
- FR004: The user should be able to update their personal details, including username, date of birth, password etc.
- FR005: The system shall provide a user dashboard.

2.2.2 Roadmap Generation

- FR001: The system shall allow users to upload a book in a pdf format.
- FR002: The user should be able to see a loading screen when waiting for book upload completion.
- FR003: The system shall analyse the uploaded content and clean it.
- FR004: The system shall be able to identify main topics in the book for roadmap generation.
- FR005: The system shall visualize the roadmap.
- FR006: The user should be able to view the roadmap generated from the book.
- FR007: The user shall be able to select a topic from the roadmap.
- FR008: The user should be able to view the progress of their generated roadmap.

2.2.3 Reels Generation

- FR001: The system shall be able to summarize the text in the topic selected by the user.
- FR002: The system shall present the user with an option to select a textual summary or reels.

- FR003: The user should be able to select either a textual summary or reels.
- FR004: The system shall be able to generate short video reels from the topic selected by the user.
- FR005: The user should be able to see generated reels based on the selected topic.
- FR006: The user should be able to provide feedback on the viewed reels.
- FR007: The system shall store the feedback.

2.3 Non-Functional Requirements

2.3.1 Reliability

1. The system should have a Mean Time Between Failures (MTBF) of at least 500 hours during normal usage to ensure minimal interruptions.
2. In case of a failure of a subsystem (e.g., text summarization or reels generation), the entire system should not fail, and fallback mechanisms should be in place to notify the users of a failure/error
3. For a system failure, the system will log necessary errors and critical issues for debugging and bug/error resolution.

2.3.2 Usability

1. **USE-1:** The user interface should be intuitive enough for first-time users to easily navigate the key features with minimal assistance.
2. **USE-2:** The system shall allow users to upload a book in utmost two interactions (e.g., selecting the file and confirming the upload).
3. **USE-3:** The system shall provide progress bars for all lengthy processes such as uploading a book or analyzing content.
4. **USE-4:** The system shall provide an interactive roadmap allowing users to navigate through the roadmap to view topics and sub-topics.
5. **USE-5:** The system shall enable users to recover from errors by providing error messages with clear description.

2.3.3 Performance

- **PER-1:** The system shall allow users to upload a book within 30 seconds, for a size up to 50 MB.
- **PER-2:** The system shall generate a personalized roadmap, within 15 seconds after course book has been successfully uploaded.
- **PER-3:** The user dashboard shall load within 3 seconds after successful user login.
- **PER-4:** The system shall display the topics and subtopics within 3 sec after clicking on the generated roadmap.
- **PER-5:** The system should be able to handle at-least 100 concurrent users without any performance issues.
- **PER-6:** The average response time for any standard operations should not be more than 3 seconds.

2.3.4 Security

- **SEC-1:** The system shall ensure secure authorized login for all registered users.
- **SEC-2:** The system must guarantee data integrity for all users, preventing any unauthorized data access.
- **SEC-3:** The system shall implement security measures that require a minimum of two levels of authentication for accessing sensitive user data.

2.4 Domain Model

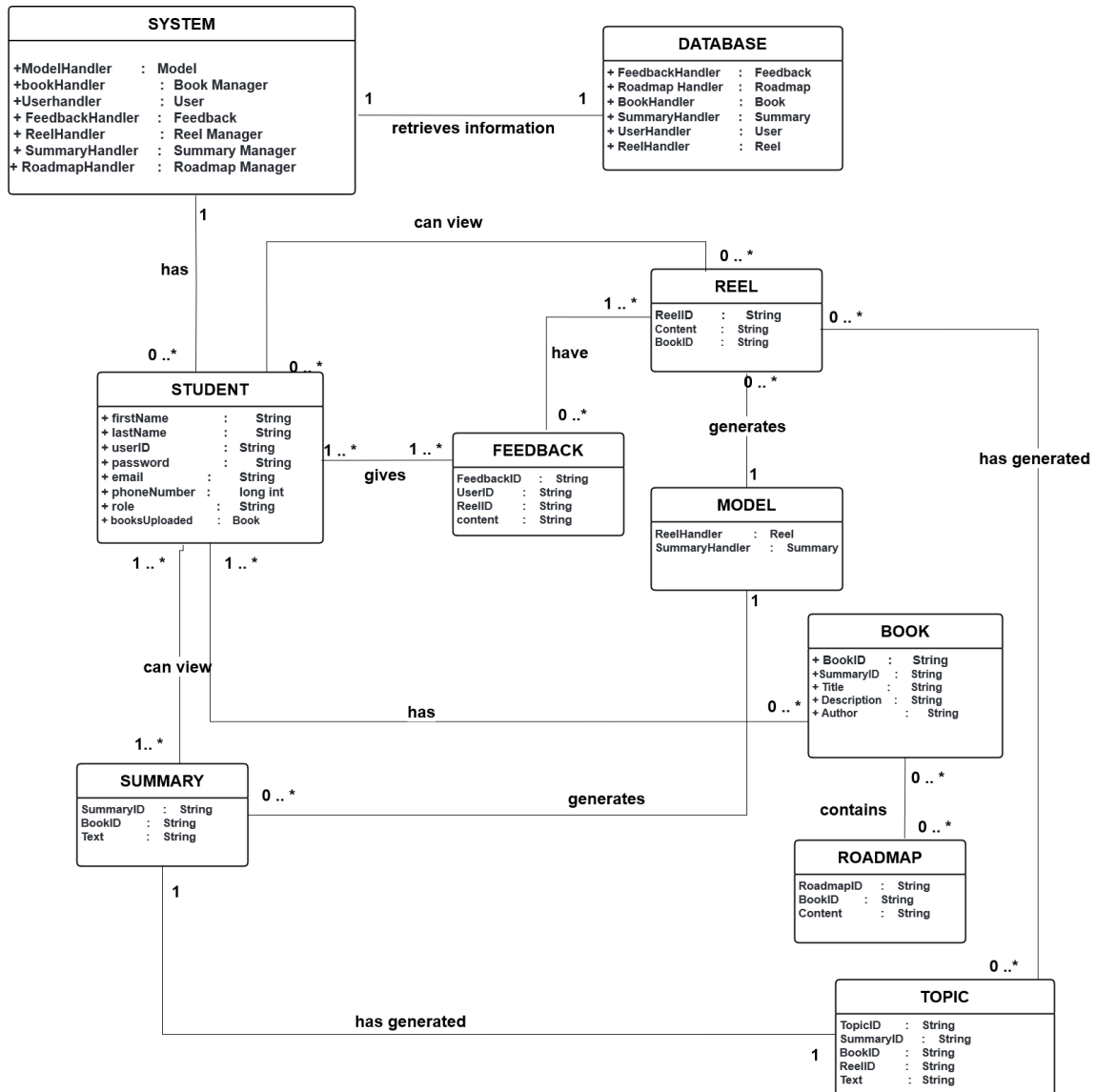


Figure 2.2: Domain Model

Chapter 3

System Overview

Scholarly is an AI Driven Smart Learning Assistant designed and developed to boost student's learning process through concise and topic-specific short reels instead of long video lectures. The primary objective of this system is to solve the problem of students being unable to retain focus while learning from traditional textbooks. Scholarly mainly functions by allowing its users to upload a book and automatically generate a roadmap to act as a study plan for the user allowing them to keep a track of their progress as they move forward. The generated roadmap will feature textual summaries as well as short video reels categorized into topics and subtopics based on the uploaded content. The design concept of this application is based on three key principles: user-friendliness, accessibility, and efficiency; mainly focusing on providing a simple yet intuitive experience for the user. The integration of AI algorithms will ensure that the system adapts to the needs of every student to offer personalized roadmaps and recommendations based on user performance and preferences.

3.1 Architectural Design

The Scholarly mobile application is designed using a three-tiered architecture, which includes the Presentation Tier, Application Logic Tier, and Data Tier. This three-tiered architecture ensures that all the functionalities and responsibilities are distributed amongst different tiers, resulting in the development of a scalable, maintainable and efficient mobile application.

1. **Presentation Tier:** The responsibilities of this tier include handling the user interface where students can upload and view content of the mobile application.
2. **Application Logic Tier:** This tier contains the core logic and functionality of the application. It includes modules for user management, content upload and extrac-

tion, roadmap generation, textual summaries generation and reels generation.

3. **Data Tier:** This Data tier of this application handles the storage, retrieval, and data management operations. It includes the database server where all user data, generated roadmaps, generated summarized text, user feedback, and other information will be stored.

3.1.1 Relationships Between Tiers

1. **Presentation Tier and Application Logic Tier:** The Presentation tier containing the mobile app interfaces communicate with the application logic tier via APIs. The interaction between both the tiers allows users to securely log in, view personal information, upload books, generate textual summaries and roadmaps etc.
2. **Application Logic Tier and Data Tier:** The Application tier contains the business logic modules of this application and can interact with the data tier to store and retrieve user data using JSON (firebase). This interaction enable the user the efficiently manage and update their data.

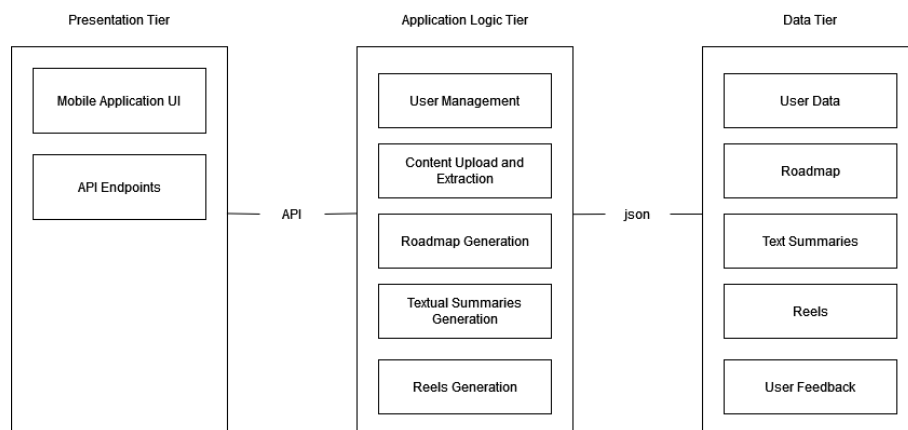


Figure 3.1: Architectural Design

3.2 Design Models

3.2.1 Activity Diagram

3.2.1.1 User Management

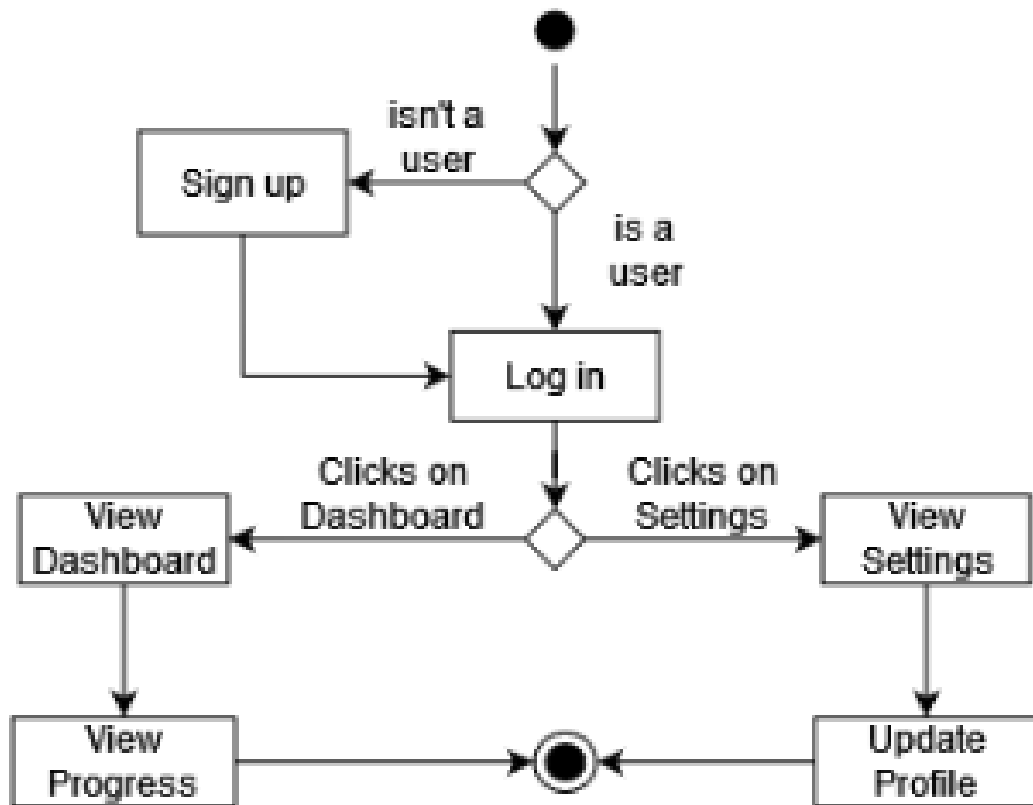


Figure 3.2: Activity Diagram

This activity diagram showcases the user creation and authentication sequence and the user settings and dashboard where if a student is already a user then they can simply log in, otherwise they can sign up and login. Afterward, if they select the dashboard, then they will view their personalized user dashboard and view their progress. If they select the settings, then they will view the settings page and can update their profile as well.

3.2.1.2 Roadmap Generation

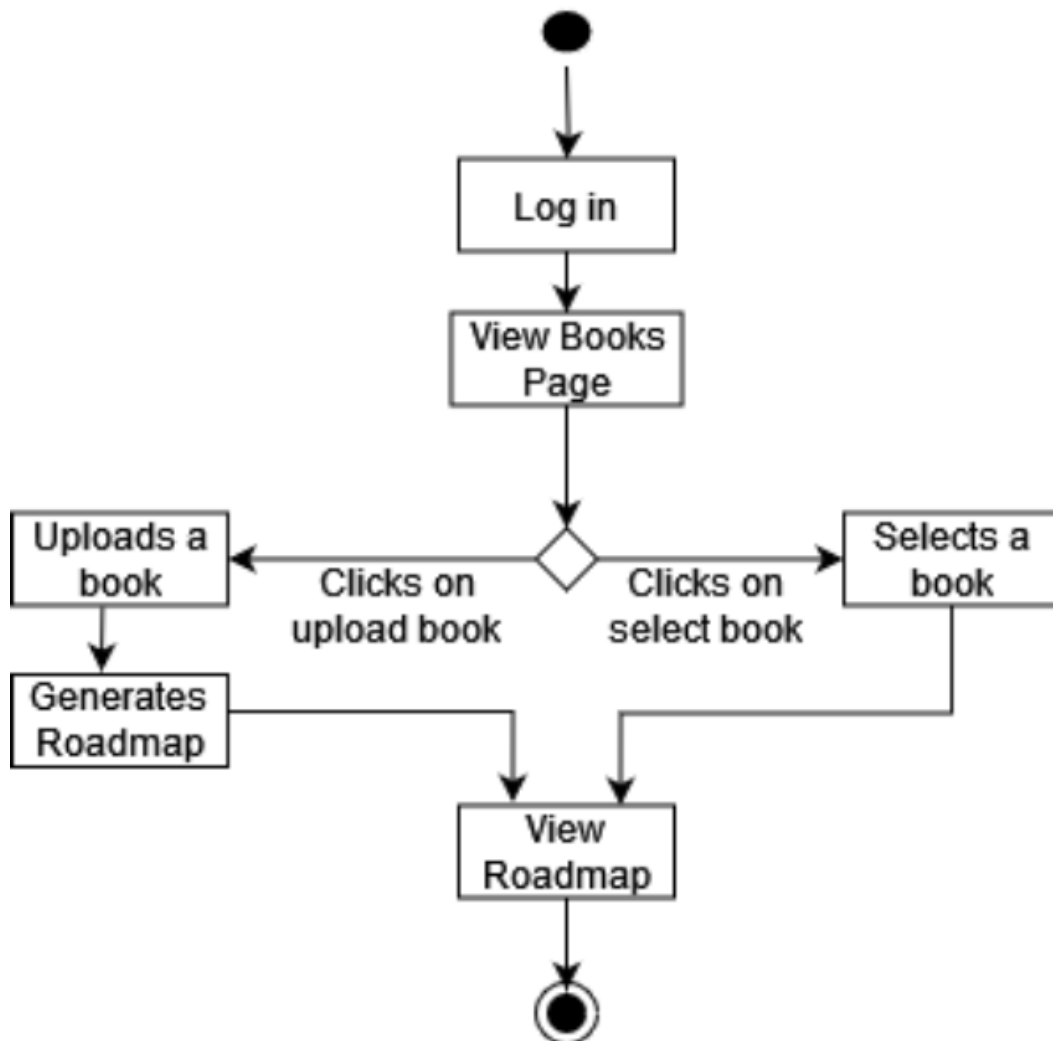


Figure 3.3: Activity Diagram

This activity diagram showcases the upload book and view/generate roadmap sequence. When a user logs in, they can view their books page and have the option to either upload a book or select an already uploaded book. If they select a book, then they will see the upload book page where they will upload a book and then a loading screen where a roadmap is being generated. Lastly, a roadmap will be displayed. If they select an already uploaded book then they will directly view the roadmap.

3.2.1.3 Reels Generation

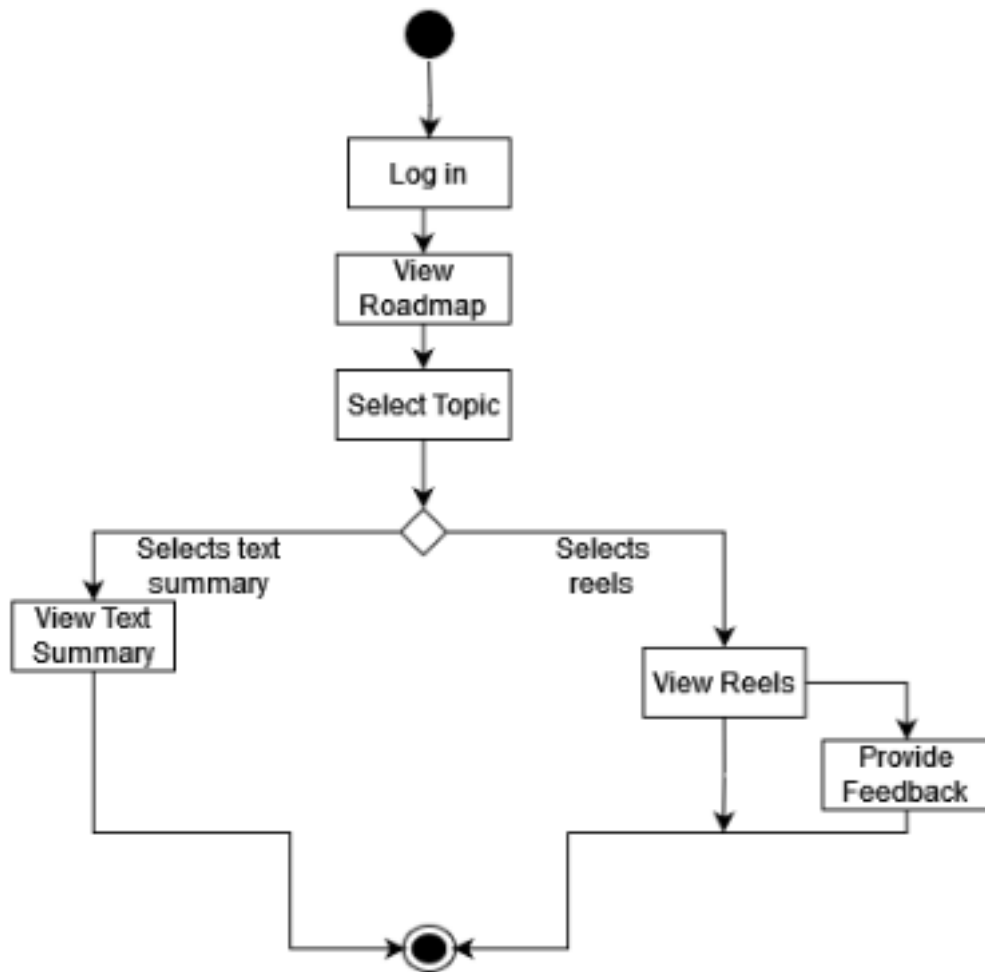


Figure 3.4: Activity Diagram

This activity diagram showcases the view reels sequence. When the user logs in and views a roadmap, they will be able to select a topic from the roadmap. Based on that topic, they can either select the text summary option or the reels option. In the text summary option, they will view a textual summary of their selected topic and if they select the reels option then they will be able to view reels for the specified topic and will also be able to provide feedback on the viewed reel.

Feedback Manager, Summary Manager, Reel Manager, and Roadmap Manager to manage the core functionality of the app.

- **User** class stores all the personal information of the such as user's first and last name, password, email, phone number, and role. It also contains methods for registering and logging in for the users.
- **Student** class inherits from the User class and implements some additional attributes such as uploadBook and giveFeedback methods to allow a student to upload a new book or provide their feedback.
- **Book** class stores the information related to an uploaded book using the BookID, Title, Description, and Author attributes.
- **Summary** class stores information for generated summaries by using SummaryID, Text, and Topic attributes
- **Reel** class holds attributes like ReelID and Content. It also manages the creation of short video explanations (reels) from the provided content.
- **Feedback** class records the provided feedback by the user for a specific reel. Attributes of this class include FeedbackID and Content.
- **Model** handles the communication with machine learning models used for generating reels and summaries. It has methods for initializing the model and generating content.
- **Roadmap, Summary, Reel, and Feedback Managers** are responsible for managing the respective functionalities and interacting with the controller class.
- **Database** interacts with all the major components and handles the data store and retrieve operations related to feedback, books, reels, summaries, and roadmaps.

3.2.3 Class-level Sequence Diagram

3.2.3.1 User Creation

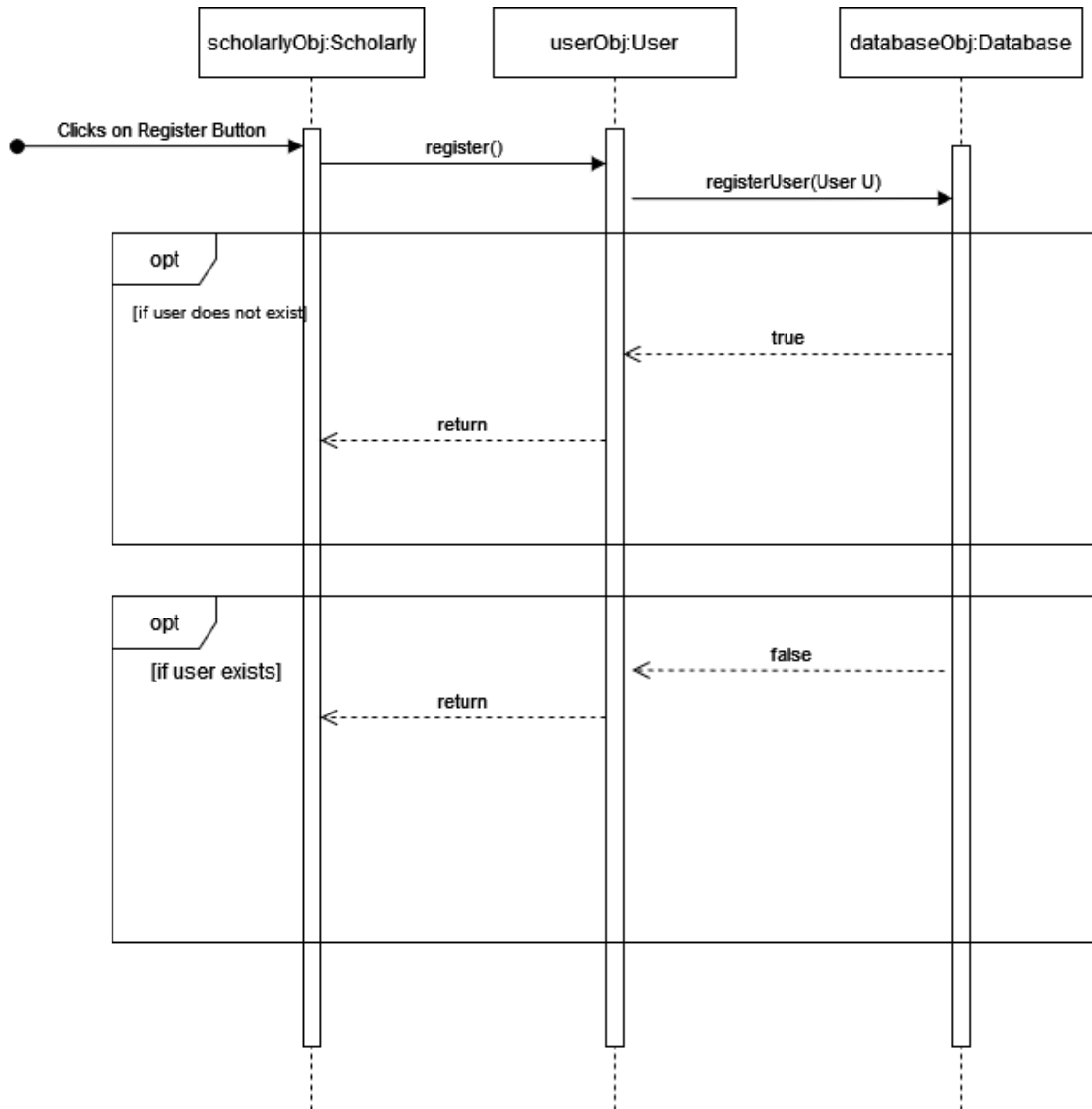


Figure 3.6: SD User Creation

This sequence diagram showcases the registration process. When the user clicks on the register button, the system class; Scholarly, calls the `registerUser()` function from the User class which requests to add a new user in Database. Given the user information, if the user does not exist then the Database adds a new user and returns true but if a user does exist, the Database returns false.

3.2.3.2 Upload Book

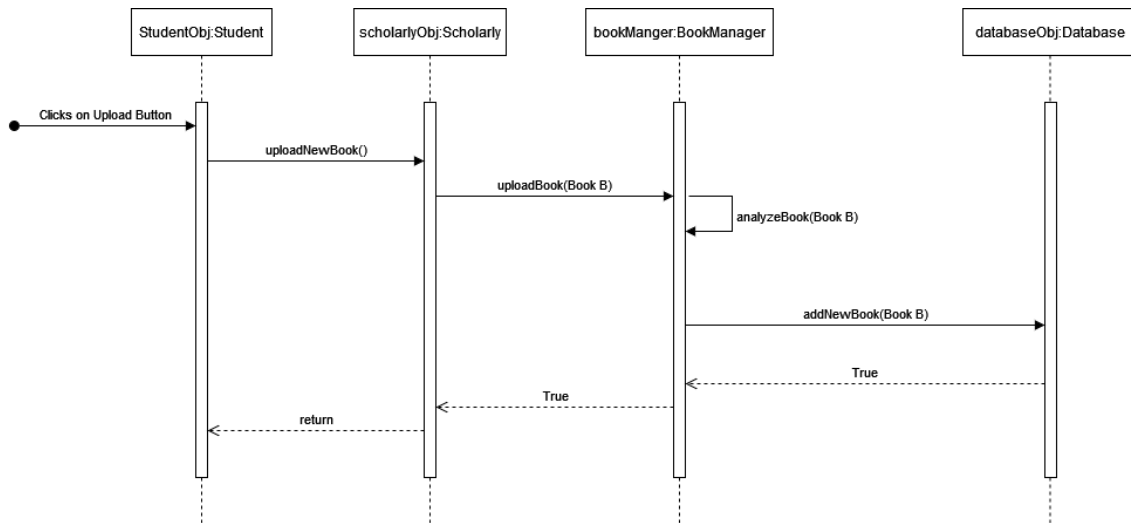


Figure 3.7: SD Upload Book

This sequence diagram showcases the book upload process. When the user clicks on the upload book button, the student class; `Student`, calls the `uploadBook()` function in system class; `Scholarly`. The system class calls a similar function in `BookManager` class which preprocess the book and adds the book into the Database.

3.2.3.3 Roadmap Generation

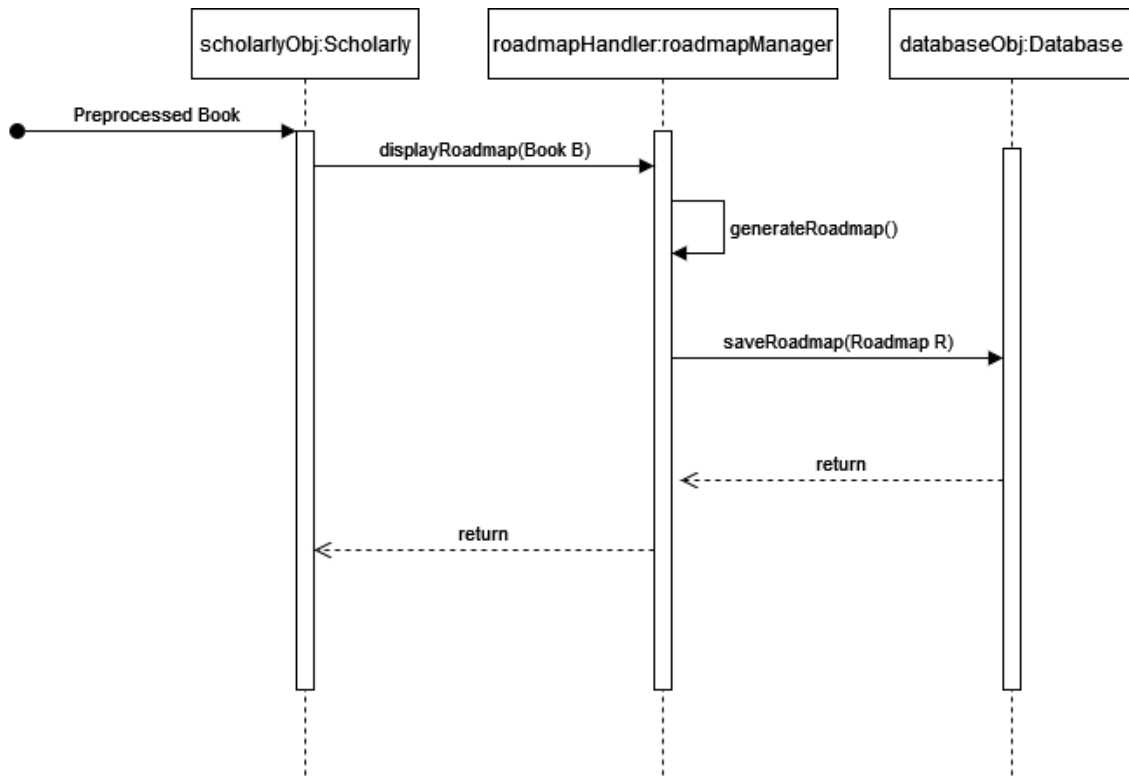


Figure 3.8: SD Roadmap Generation

This sequence diagram showcases the roadmap generation process. When the book is uploaded, the preprocessed book is sent to the system class; Scholarly, which calls the `displayRoadmap()` function in the `roadmapManager` class. This class then generates a roadmap and loads it into the Database and returns it to the system class.

3.2.3.4 Generate Topic Summary

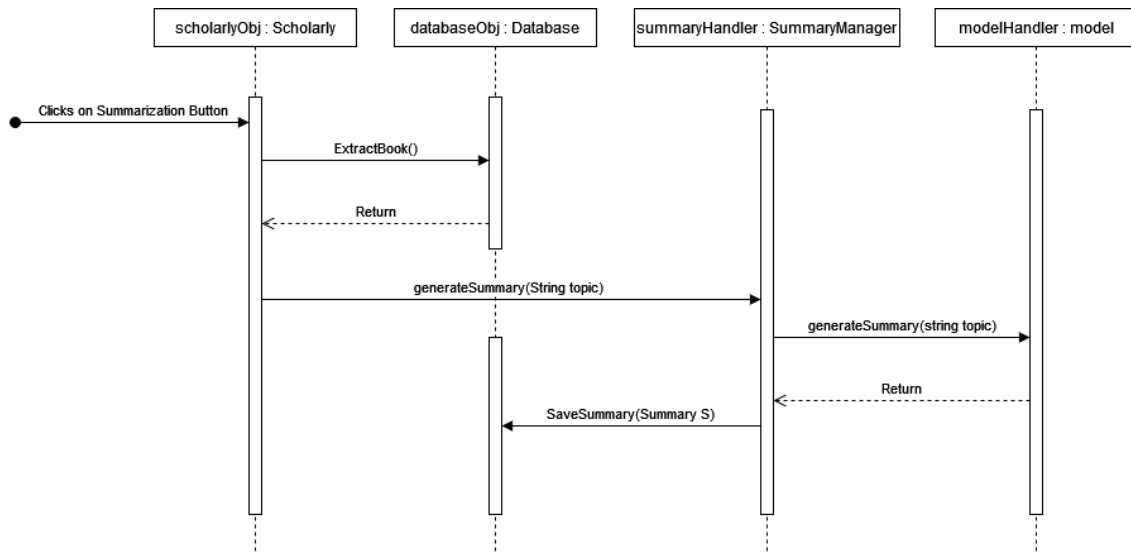


Figure 3.9: SD Generate Topic Summary

This sequence diagram showcases the text summarization process. When the user clicks on the summarize button, the system class; Scholarly, extracts the book from the Database. It sends this book to the SummaryHandler which interacts with the Model to generate a summary and then saves the summary into the Database.

3.2.3.5 View Topic Summary

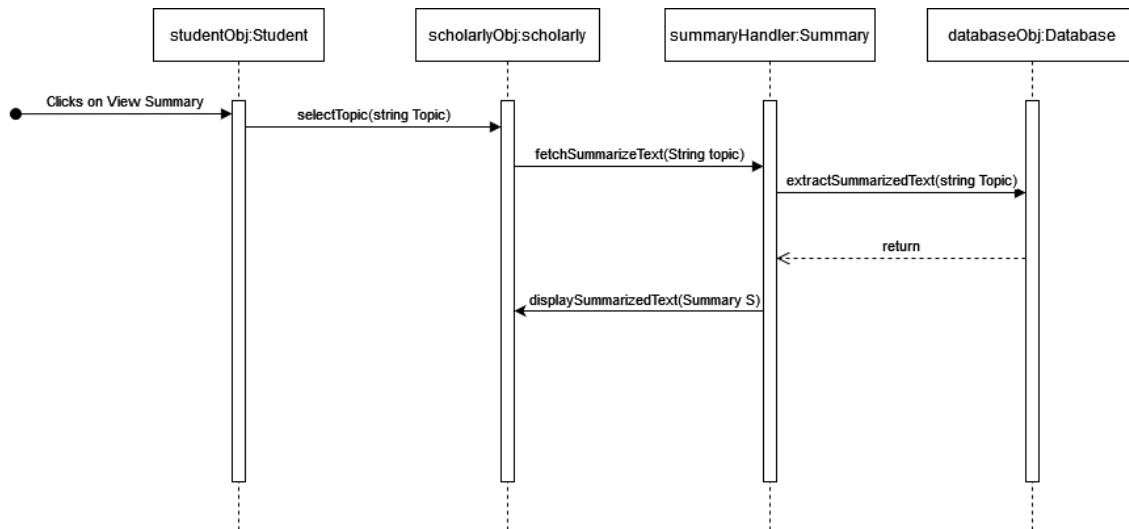


Figure 3.10: SD View Topic Summary

This sequence diagram showcases the view summary process. When the user clicks on view summary and they select the topic, the system class; Scholarly, fetches the summarized text from the Summary class. The Summary class extracts the summarized text from the Database and returns it. The system class then displays the summary.

3.2.3.6 Generate Reels

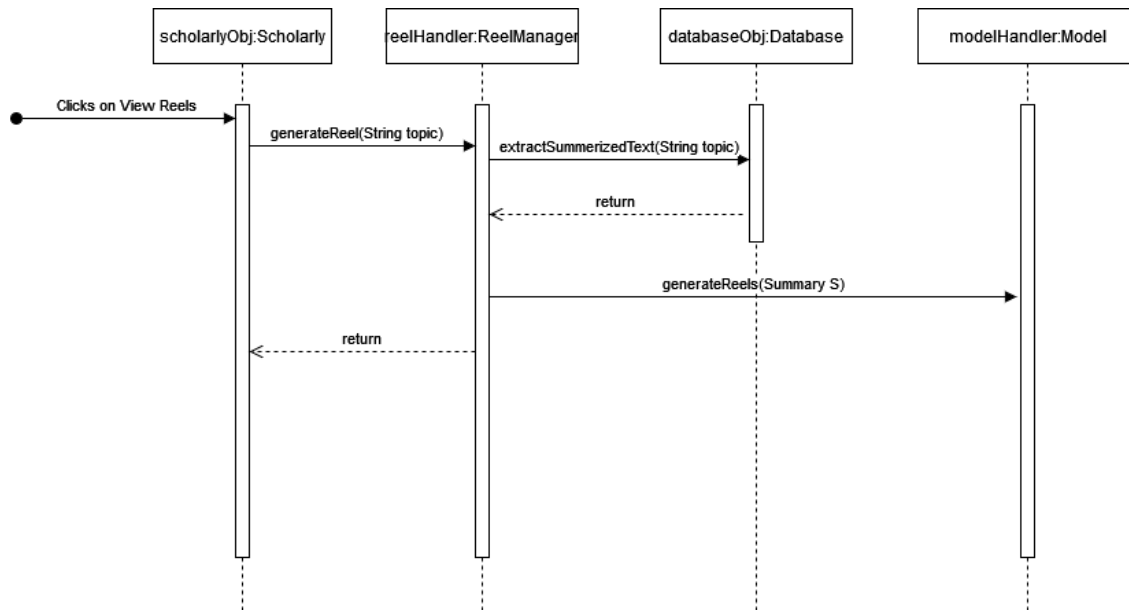


Figure 3.11: SD Generate Reels

This sequence diagram showcases the reel generation process. When the user clicks on view reels button, the system class; Scholarly, calls the `generateReel()` function in the `ReelManager` class. The `ReelManager` first extracts the summarized text from the Database and then sends it to the Model to generate the reels. The reels are then returned to the `ReelManager` class.

3.2.3.7 View Reels

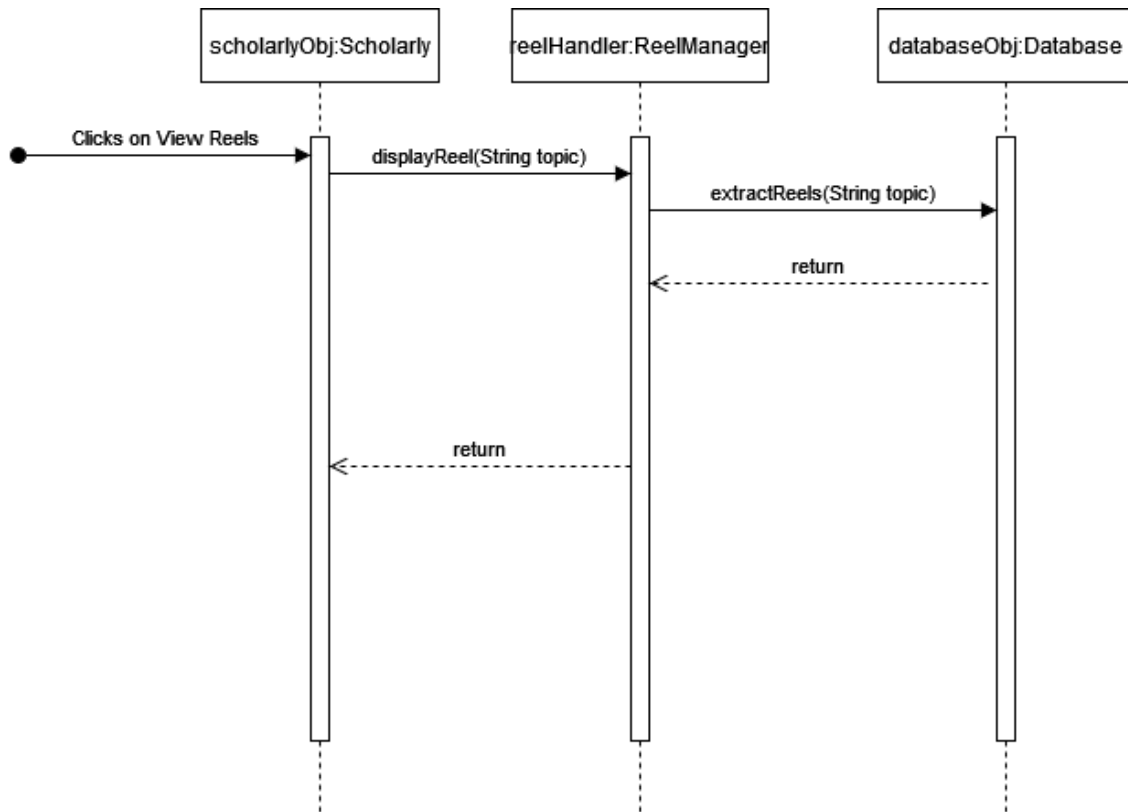


Figure 3.12: SD View Reels

This sequence diagram showcases the view reels process. When the user clicks on the view reels button, the system class; Scholarly, calls the `displayReel()` function in the `ReelManager` class. This class then extracts the reels from the `Database` and displays it.

3.2.3.8 Provide Feedback

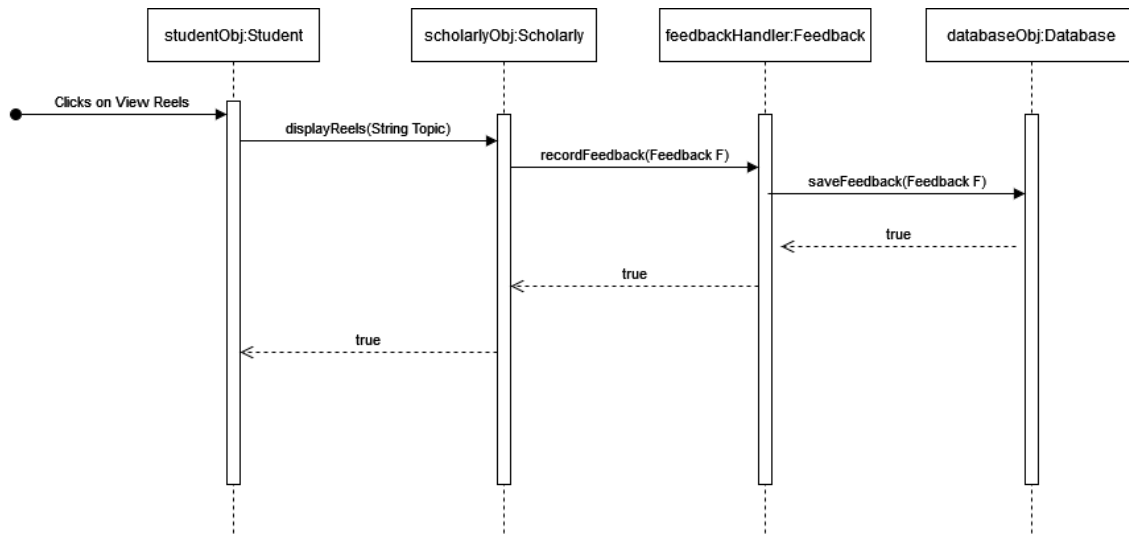


Figure 3.13: SD Provide Feedback

This sequence diagram showcases the feedback process. When the user has clicked on the view reels button and the reels are displayed, the system class; Scholarly, calls the record-Feedback() function in the Feedback class which saves the feedback into the Database.

3.3 Data Design

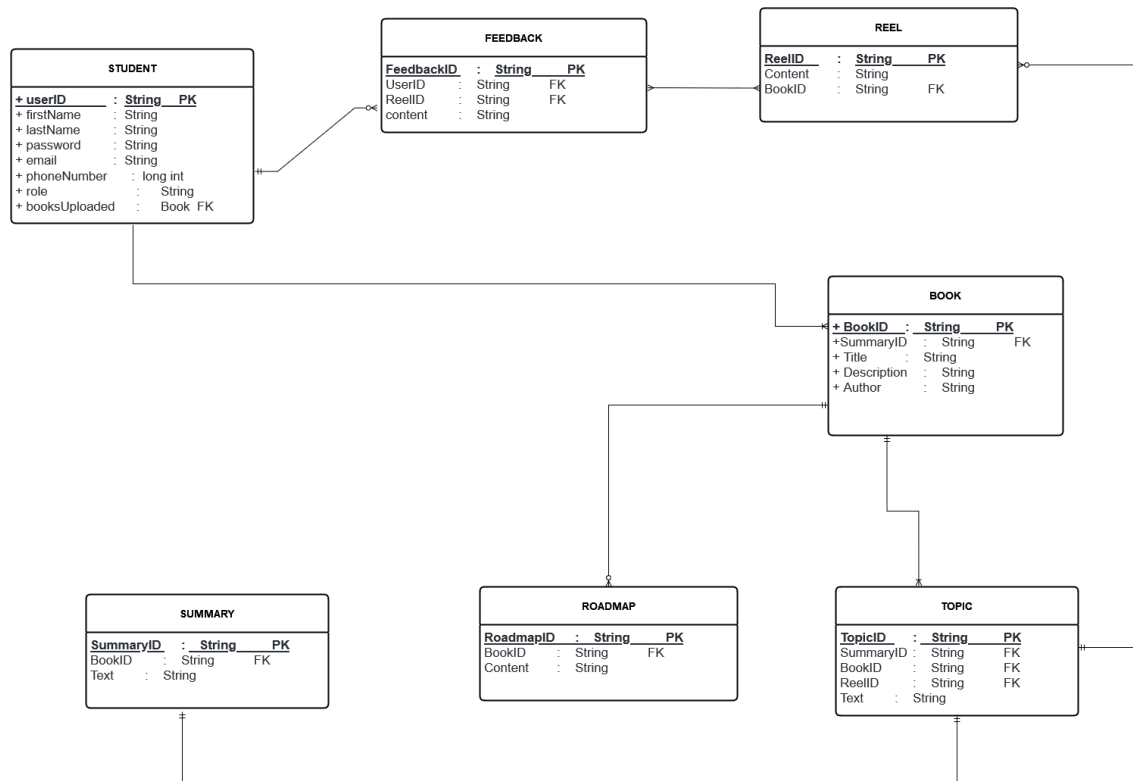


Figure 3.14: Data Design Diagram

The data design of the Scholarly app translates the system's information into well-structured data models which are stored and managed using a database system to support core functionalities such as user management, content processing, feedback, summaries, and roadmaps. Each entity in the data design is defined with clear relationships and attributes, ensuring smooth data interactions throughout the application.

3.3.1 Data Structures

3.3.1.1 Student

- **Attributes:** `userID` (PK), `firstName`, `lastName`, `password`, `email`, `phoneNumber`, `role`, `booksUploaded` (FK).
- **Description:** Stores essential student information and keeps a record of all the books uploaded by each student.

3.3.1.2 Book

- **Attributes:** BookID (PK), SummaryID (FK), Title, Description, Author.
- **Description:** Stores book information for a book uploaded by the user.

3.3.1.3 Summary

- **Attributes:** SummaryID (PK), BookID (FK), Text.
- **Description:** Stores the Textual Summaries for each topic and subtopic of the uploaded book

3.3.1.4 Reel

- **Attributes:** ReelID (PK), Content, BookID (FK).
- **Description:** Stores short-form videos (reels) for each topic, generated based on the uploaded book content.

3.3.1.5 Roadmap

- **Attributes:** RoadmapID (PK), BookID (FK), Content.
- **Description:** Stores the generated roadmap for the each uploaded book.

3.3.1.6 Feedback

- **Attributes:** FeedbackID (PK), UserID (FK), ReelID (FK), Content.
- **Description:** Contains the feedback provided by students for a specific reels

3.3.1.7 Topic

- **Attributes:** TopicID (PK), SummaryID (FK), BookID (FK), ReelID (FK), Text.
- **Description:** Contains the topic of the uploaded book along with the ids for generated reel and summary.

3.3.2 Databases and Storage

- **Firebase:** Will be used to store personal data such as student information as well as uploaded/generated content including books, summaries, reels, feedback, and roadmaps.

Bibliography

- [1] Neil A. Bradbury. Attention span during lectures: 8 seconds, 10 minutes, or more?
Advances in Physiology Education, 41(4):509–513, 2016.